THE WARM RETURN 京张回暖线

A0 board 1 / 2

Giving the heat of compute back to the city: Zhangbei green power → Haidian compute → urban waste heat | one spine, ten nodes, three stitches, one blank

The design argument

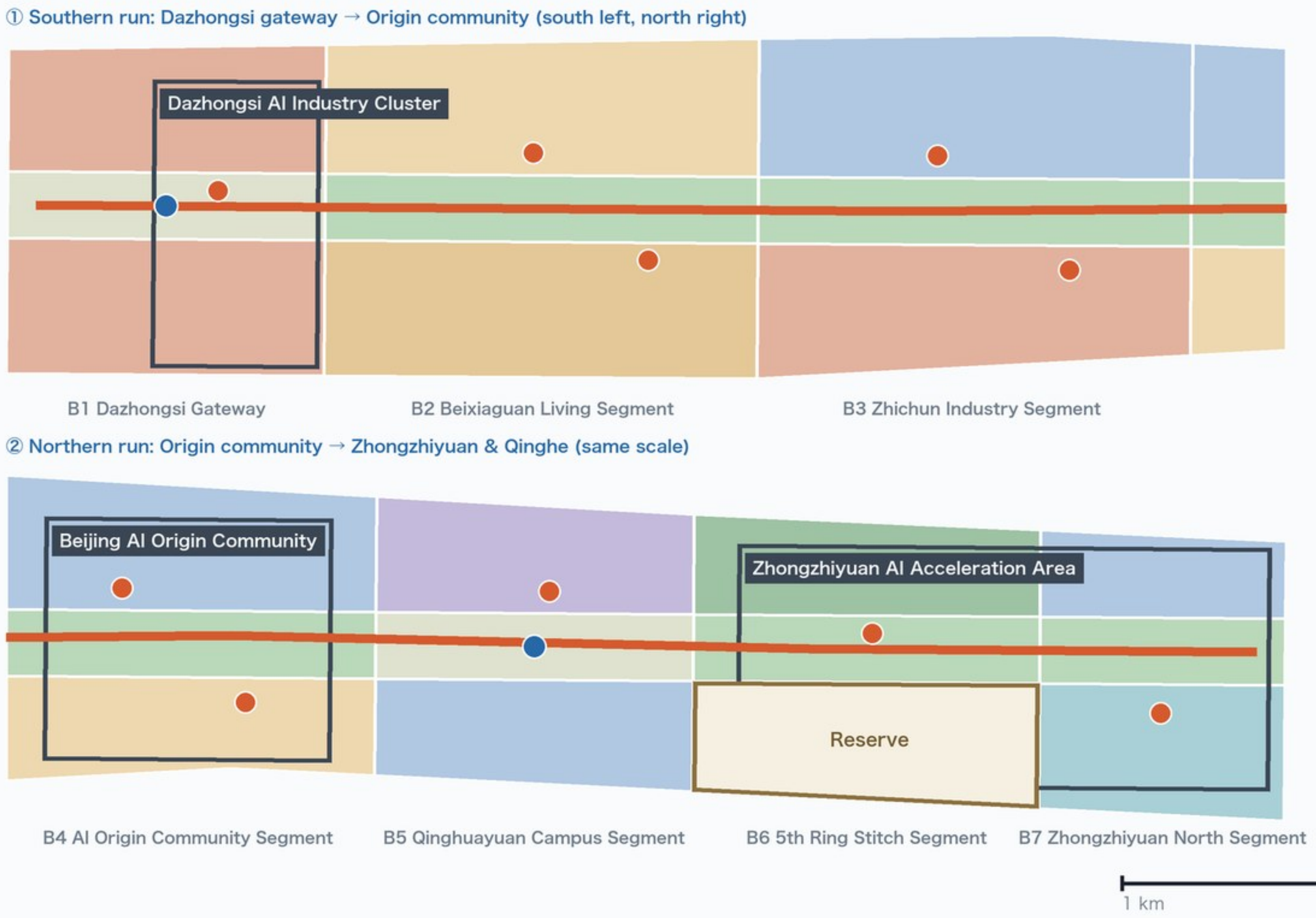
The first branch of the Jing-Zhang railway was built to carry coal; a hundred and twenty years later the same corridor carries compute. Green power flows east, compute flows east, and the heat mostly dissipates in cooling towers. This proposal takes the return leg of that energy chain — heat coming back to the city — as the organising motif of a nine-kilometre heritage park, and draws only what can be verified.

Overall design area	Key areas	Green + public space	Slow-mobility share	Registered sources
1,140.0 ha	**368.4 ha**	**26.67%**	**86.8%**	**33**
announced	announced · 3	recomputed	recomputed	with URL and licence

Overall Spatial Structure and Land-Use Layout

Fig. 02 / Land use

The Warm Return · one spine, three anchors, ten nodes, one reserved parcel | 21 seamless land-use cells, coverage 0.999993



Land-use classes (MNR classification guide)

- 05 Commercial & business — 200.9 ha
- 07 Residential — 148.4 ha
- 0702 Community service facilities — 86.8 ha
- 0802 Research & development — 250.0 ha
- 0804 Education — 50.6 ha
- 0805 Sports — 47.3 ha
- 1401 Park green space — 186.4 ha
- 1402 Buffer green space — 47.9 ha
- 1403 Public square — 62.6 ha
- 16 Reserved (blank) land — 60.4 ha

Structural elements

- Warm Return spine 9,494 m
- Warm public nodes ×10
- Station-city plazas ×2
- Three provisional key areas

The nine-kilometre corridor is cut into two runs at one shared scale; the seam between runs is not a spatial break.

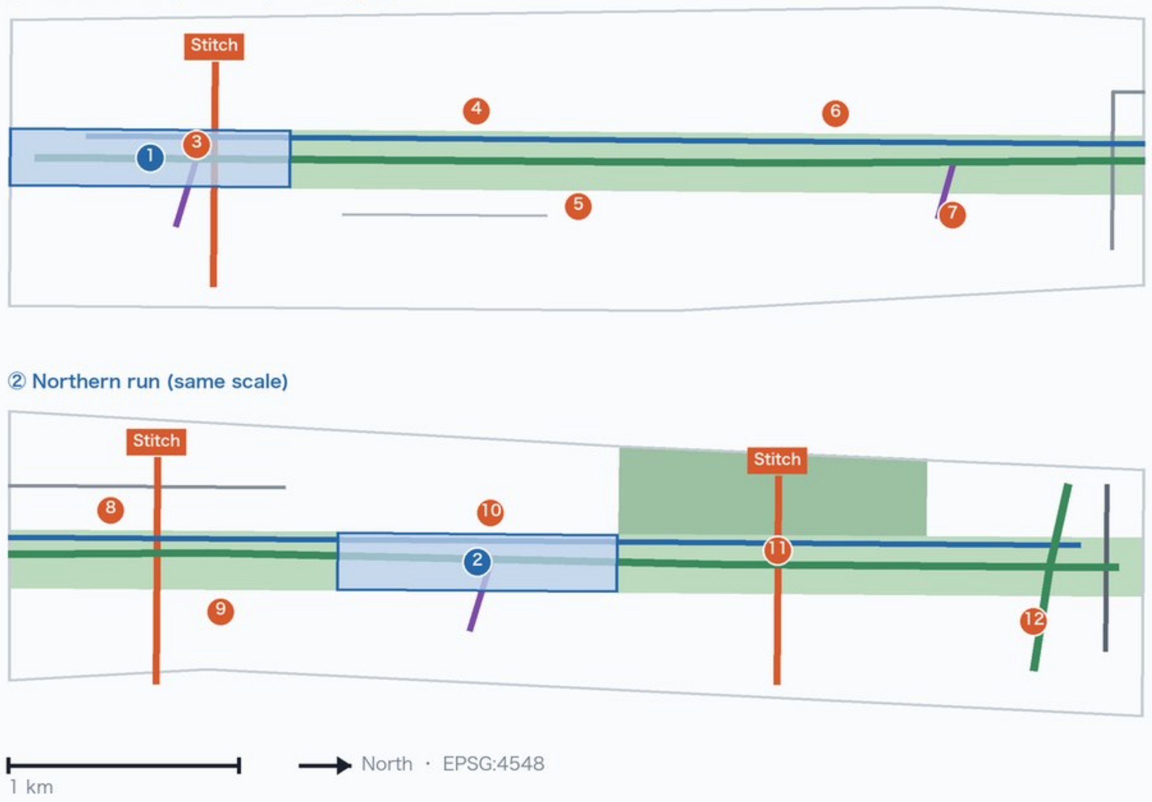
→ North · EPSG:4548

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released.

Slow Mobility and Blue-Green Public Space

Fig. 04 / Mobility & blue-green

One spine, three stitches, three transit connectors | slow mobility is 86.8% of network length; green and public space together 26.7%



Network hierarchy

- Greenway / slow-mobility spine — 10,319 m
- Continuous cycleway — 9,105 m
- Pedestrian stitch — 2,862 m
- Transit connector — 872 m
- Secondary road — 726 m
- Micro-circulation branch — 1,905 m
- Local access road — 888 m

Green and public space

- Park green space 1401 ×5 — 186.4 ha
- Buffer green space 1402 ×1 — 47.9 ha
- Station-city plazas ×2 — 62.6 ha
- Warm public nodes ×10 — 5.11 ha

Three things deliberately not drawn

This proposal does not propose rail alignment or station changes, does not estimate ridership, and does not draw heating pipe routes, diameters or connection capacities. All three depend on official data that is not public; drawing them would turn guesses into conclusions.

Public space node index

1. Dazhongsi Warm Gateway Plaza
2. Qinghuayuan Station Foyer
3. Dazhongsi Warm Room
4. Beixiaguan Warm Pavilion
5. Beixiaguan Service Point
6. Zhichun Heat Interface Hall
7. Zhichun Enterprise Lounge
8. Origin Open-Source Lounge
9. Origin Talent Lounge
10. Qinghuayuan Foyer Pavilion
11. 5th Ring Warm Gallery
12. Qinghe Warm Greenhouse

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released. Network lengths are recomputed from the submitted LineStrings in EPSG:4548 and exclude carriageway width or area.

Four things that can be drawn, computed and checked

One spine: it runs the full nine kilometres; 23.16 km of slow mobility network-wide.

Three stitches: they address where the corridor is severed, crossing rather than detouring.

Ten nodes: warm public nodes total 5.1 ha, giving winter public life somewhere to stay.

One blank: 60.4 ha of reserved land keeps a heat interface possible; no unfounded pipeline today.

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Key Areas, Recomputation Protocol and Refusal List

Three key areas announced at 368.4 ha | 47 of 57 metrics recomputed, 10 left unknown | 25 self-checks, 0 blocking

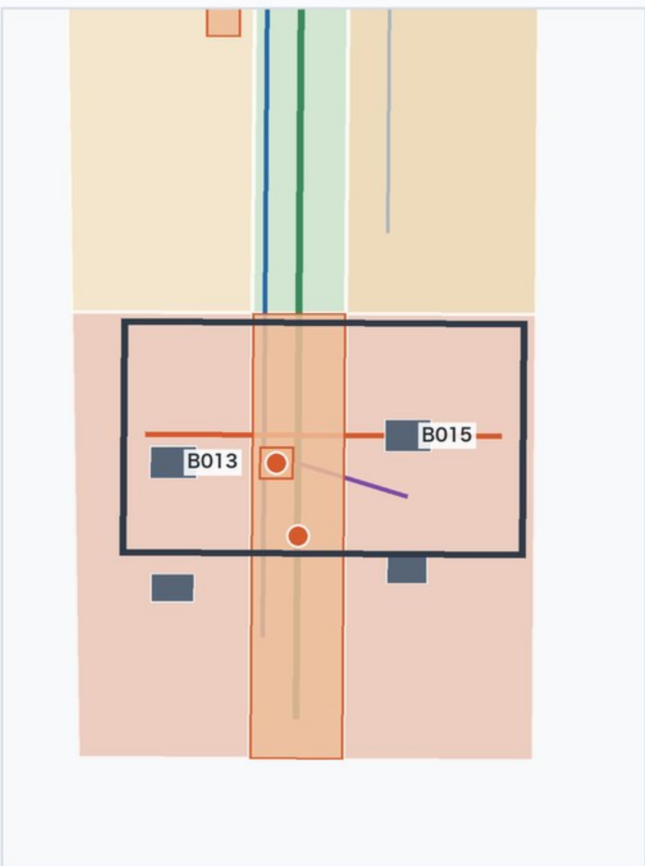
Three Key Areas: Detailed Design Index

Fig. 03 / Key areas

Provisional extents of the three announced key areas with this proposal's detailed design elements | all three panels share one scale, so areas and distances compare directly

1 | Dazhongsi AI Industry Cluster

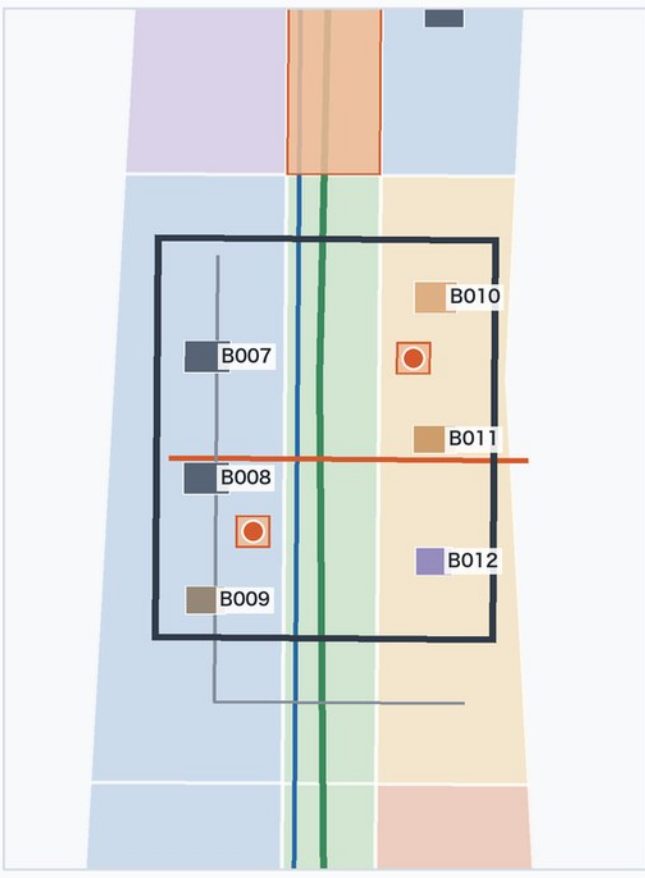
Announced ≈72.0 ha · recomputed 72.0 ha · rough inference



The city end: the rail gateway meets the commercial frontage; the station-city plaza hands metro flows straight to the Warm Return spine.

2 | Beijing AI Origin Community

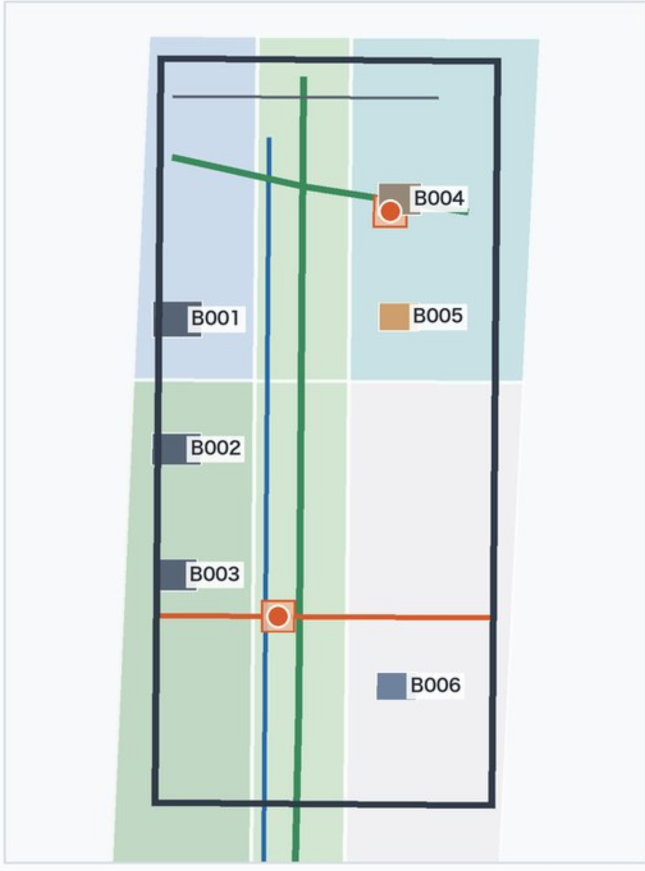
Announced ≈104.3 ha · recomputed 104.3 ha · rough inference



The human end: talent housing, incubation and campus-city links concentrate here; two warm lounges carry everyday dwelling and a micro-circulation branch carries arrival.

3 | Zhongzhiyuan AI Acceleration Area

Announced ≈192.1 ha · recomputed 192.9 ha · rough inference



Supply end: compute and heat sources sit here, so public space is closest to the technical plant. The reserved parcel adjoins it, holding room for heat exchange and municipal interfaces.

New or renewed mass (concept) · Retained buildings ×3 · Warm nodes and plazas · Greenway and spine · Pedestrian stitch · Transit connectors

500 m

North · EPSG:4548

Boundaries and layers are AI-generated conceptual proposals derived from the public announcement and the repository's provisional geometry. They are not an official redline and must be recomputed once official boundaries are released. Building masses, heights and intensities are indicative only; control indicators await the statutory plan.

Metric Recomputation and Evidence Chain

Fig. 05 / Metrics

57 metrics in three classes: 47 recomputed from the submitted geometry, 8 awaiting official controls, 2 awaiting operational data | the latter 10 stay unknown, never estimated

The three metric classes

class 1 recomputable	47
recomputed from GeoJSON in EPSG:4548	
class 2 pending official control	8
needs an approved control plan	
class 3 operational performance	2
needs IT-load and heat-network data	

Confidence distribution (57 metrics)

high	10
medium	31
low	6
unknown	10

Low confidence clusters in building masses and key-area sizes: both derive from provisional geometry.

How anyone can independently recompute

① Read the nine GeoJSON layers in geometry/ (EPSG:4326, lon-lat order).

② Project to EPSG:4548 before computing area and length; never compute area in 4326.

③ Compare each result with the formula field in metrics.json and reproduce the same number.

④ A deviation above 0.5% counts as an error in this package; please raise it in PR review.

Main data files behind the metrics

geometry/land_use.geojson	12
brief/site-package/ranges/planning_limits.json	10
geometry/site_boundary.geojson	7
geometry/roads.geojson	7
geometry/public_space.geojson	6

class 1 recomputation sample (13 of 47)

Metric	Formula	Source	Value
site_area_sqm	polygon_area(site_boundary, EPSG:4548)	site_boundary.geojson	1,141.3 ha
official_overall_design_area_sqm	official_announcement_value(overall_design_area)	planning_limits.json	1,140.0 ha
site_area_deviation_from_official_ratio	abs(site_area_sqm - official_overall_design_area_sqm) / official_overall_d	site_boundary.geojson, planning_limits.json	0.11%
land_use_coverage_ratio	area(intersection(union(land_use), site_boundary)) / site_area_sqm	land_use.geojson, site_boundary.geojson	100.00%
land_use_area_0802_sqm	sum(polygon_area(land_use where land_use_code=0802))	land_use.geojson	250.0 ha
green_ratio	green_space_area_sqm / site_area_sqm	green_space.geojson, site_boundary.geojson	20.52%
public_space_ratio	public_space_area_sqm / site_area_sqm	public_space.geojson, site_boundary.geojson	6.14%
warm_public_node_count	count(public_space where public_space_type_zh='回暖公共节点')	public_space.geojson	10
road_centerline_length_m	sum(line_length(road_centerline, EPSG:4548))	roads.geojson	26.68 km
slow_mobility_share_of_road_network_ratio	slow_mobility_length_m / road_centerline_length_m	roads.geojson	86.80%
provisional_key_area_total_sqm	area(union(key_areas))	key_areas.geojson	369.3 ha
official_key_detailed_design_area_sqm	official_announcement_value(key_detailed_design_area)	planning_limits.json	368.4 ha
self_check_blocking_failures	count(self_check.checks where result='fail' and severity='blocking')	self_check.json	0

class 2: 8 awaiting official control

building_density
No building survey or approved control plan

total_floor_area_sqm
Needs approved FAR and storey controls

floor_area_ratio
FAR is a statutory control, not derived by a proposal

building_height_control_m
Height limits need approvals (airspace, heritage)

pending_official_green_ratio_control
Official green-ratio control still pending

setback_control_m
Setbacks only follow the official road redline

road_area_sqm
No official redline or cross-sections; no road-area polygons

road_area_ratio
Both terms depend on the official road redline

class 3: why the two heat metrics must stay blank

waste_heat_recoverable_share_of_it_load_ratio
No public data on IT capacity, cooling type or return temperature

heat_supplied_households_equivalent
Heat load and network interfaces unknown; "N households" would be pseudo-precision

Waste-heat reuse is this proposal's central claim, so these two metrics are the ones a reader most wants — and precisely the ones that must not be invented. The recoverable share depends on installed IT capacity, cooling type, return temperature and the annual load curve; the household figure adds heat-network interfaces and heat demand on top. None of the four is verifiable in public sources.

What this proposal offers instead is a measurable set of interface conditions — heat-exchange room locations, utility-corridor reservations and end-user terminals — not an unverifiable recovery rate.

This diagram asserts no spatial boundary. All values are read from metrics.json and recomputed in EPSG:4548; everything must be recomputed once official boundaries and key-area polygons are released. 'unknown' is a deliberate choice, not an omission.

Protocol: anyone can re-run it

- The nine GeoJSON layers in geometry/ are stored in EPSG:4326 (lon-lat order).
- Area and length are computed only after projecting to EPSG:4548, never in 4326.
- The formula field in metrics.json states each algorithm for line-by-line reproduction.
- A deviation above 0.5% counts as an error here; please raise it in PR review.
- Each of the 12 assumptions states its retirement condition and closes once official data arrives.

Refusal list: conclusions withheld on purpose

- The five statutory controls: FAR, density, height limit, green ratio, setback.
- Total floor area, road-land area and its share.
- Recoverable waste-heat share and households heated — the two most wanted are the two that must not be invented.
- Ridership forecasts, carbon reduction, investment estimates and land revenue.
- Precise plot redlines and any adjustment to existing rail alignment.

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